

DECISION SUPPORT SYSTEMS & DATA WAREHOUSE

CO1: DESIGN DATA WAREHOUSE ARCHITECTURES
WITH APPROPRIATE DIMENSIONAL MODELS FOR
EFFECTIVE DECISION SUPPORT SYSTEMS.

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DECISION SUPPORT SYSTEMS (DSS)

A Decision Support System is an interactive computer-based system designed to assist decision-makers by using data, models, and analytical tools. It combines data, analytical models, and software tools to solve semi-structured and unstructured business problems.

Why is DSS Important?

Improves the quality of business decisions.

Supports strategic, tactical, and operational planning.

Helps organizations respond quickly to changing business conditions.

Reduces uncertainty by providing accurate information.

TYPES OF DSS

Data-driven: Focuses on analyzing large volumes of structured data.

Model-driven: Uses analytical models to support decision-making.

Knowledge-driven: Provides expert recommendations based on stored knowledge.

Document-driven: Retrieves and analyzes information from documents.

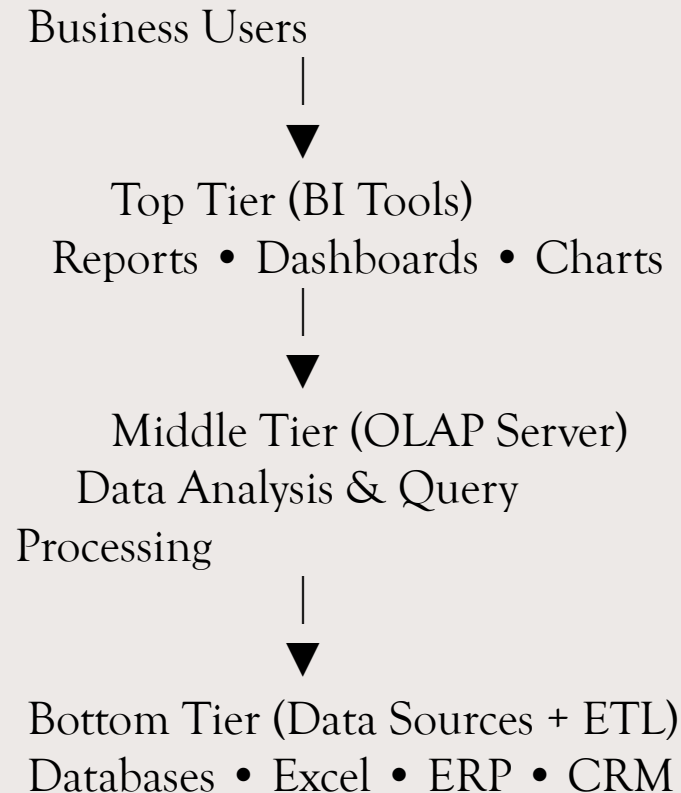
Communication-driven: Supports collaboration and group decision-making.

OPERATIONAL SYSTEMS VS DSS (OLTP VS OLAP)

Feature	OLTP (Online Transaction Processing)	OLAP (Online Analytical Processing)
Purpose	Processes daily business transactions	Analyzes data for decision-making
Data	Uses current and operational data	Uses historical and summarized data
Users	Used by many employees and customers	Used by managers and business analysts
Operations	Insert, Update, Delete transactions	Query, Analysis, Reporting

ARCHITECTURE OF DATA WAREHOUSE

Definition: A Data Warehouse is a centralized repository that stores integrated historical data.



ETL, DATA MARTS & METADATA

Extract: Collects data from multiple source systems.

Transform: Cleans, converts, and formats data for consistency.

Load: Stores the transformed data into the data warehouse.

Data Mart: A department-specific subset of a data warehouse.

Metadata: Information that describes the structure and contents of stored data.

Business Data



Data Collection



Data Processing



Analytical Models



Report Generation



Manager Decision

OLAP & DATA CUBES

OLAP: Performs multidimensional analysis to support business decisions.

Data Cube: Organizes data by Product, Time, and Location dimensions.

Operations: Slice, Dice, Roll-up, and Drill-down enable flexible data analysis.

Slice & Dice: Slice filters one dimension, while Dice analyzes multiple dimensions.

Applications: OLAP is widely used in sales analysis and business intelligence.

DIMENSIONAL DATA MODELING

Fact Table: Stores numerical measures and business transaction data.

Dimension Tables: Store descriptive data such as Customer, Product, Time, and Location.

Star Schema: Uses one fact table with multiple dimension tables for fast querying.

Snowflake Schema: Uses normalized dimension tables to reduce data redundancy.

Schema Difference: Star Schema is simpler and faster, while Snowflake Schema is more normalized and space-efficient.

QUERY PROCESSING, BI TOOLS & KNIME

Parsing: Checks the SQL query for syntax and correctness.

Optimization: Selects the most efficient execution plan for the query.

Execution: Retrieves and returns the requested data from the database.

BI Tools: Power BI, Tableau, Apache Superset, and Metabase are popular data visualization tools.

KNIME Workflow: Read Data → Transform → Analyze → Visualize → Report.

CONCLUSION

Summary:

DSS improves organizational decisions.

Data warehouses store integrated data.

OLAP enables fast analysis.

BI tools generate meaningful reports.

Future Scope:

AI-powered DSS, Cloud Data Warehousing, Real-time Analytics.